

Gravity of a Sigmoid Lopolith and Comparison to a Sphere

1. Sigmoid and Scale Parameters

We model radial smoothing with a sigmoid

$$s(r) = \frac{1}{2} \left[1 - \tanh\left(\frac{r}{w}\right) \right],$$

where w controls the transition width; as $w \rightarrow 0$, $s(r)$ approaches a Heaviside step. The derivative is

$$s'(r) = -\frac{1}{2w} \operatorname{sech}^2\left(\frac{r}{w}\right).$$

Throughout, $J_0(\cdot)$ denotes the Bessel function of the first kind (order zero).

2. Lopolith Geometry (Model B)

Let the top and bottom surfaces be

$$a_{\text{top}}(r) = z_0 + A_{\text{top}} s(r), \quad a_{\text{bot}}(r) = z_0 - A_{\text{bot}} s(r),$$

so the vertical thickness is

$$t(r) = a_{\text{top}}(r) - a_{\text{bot}}(r) = (A_{\text{top}} + A_{\text{bot}}) s(r).$$

Special cases:

- Flat top, curved base: $A_{\text{top}} = 0$.
- Curved top, flat base: $A_{\text{bot}} = 0$.
- Symmetric lens (lopolith): $A_{\text{top}} = A_{\text{bot}}$ (positive for a lens, negative for a trough).

3. Areal Mass and the Hankel Transform (intuition and formula)

Assuming constant density contrast $\Delta\rho$, the areal mass (thin-sheet approximation) is

$$\Delta m(r) = \Delta\rho t(r).$$

Why J_0 appears. An axisymmetric sheet can be decomposed into rings of radius r and width dr . The vertical gravity at observation radius R from such a ring, integrated over azimuth, reduces in the wavenumber domain to a product with $J_0(kR)$ by the Bessel addition theorem. This is the axisymmetric analogue of using the 2-D Fourier transform for cartesian symmetry.

Hankel transform pair. We use the zeroth-order Hankel transform

$$\tilde{f}(k) = \int_0^\infty f(r) J_0(kr) r \, dr, \quad f(r) = \int_0^\infty \tilde{f}(k) J_0(kr) k \, dk.$$

Vertical gravity as a spectral low-pass. For a reference depth z_0 ,

$$g_z(R) = 2\pi G \int_0^\infty J_0(kR) e^{-kz_0} k \widetilde{\Delta m}(k) dk.$$

The factor e^{-kz_0} is a low-pass: deeper sources are smoother. For our sigmoid, using the known transform of sech^2 and integrating once with respect to r , we have the accurate closed-form approximation

$$\widetilde{s}(k) \approx \frac{\pi w^2}{2} \frac{k w}{\sinh\left(\frac{\pi k w}{2}\right)}.$$

Hence

$$\widetilde{\Delta m}(k) \approx \Delta \rho (A_{\text{top}} + A_{\text{bot}}) \frac{\pi w^2}{2} \frac{k w}{\sinh\left(\frac{\pi k w}{2}\right)}.$$

4. Vertical Gravity Anomaly: Exact Integral and Analytic Approximation

Exact Hankel integral.

$$g_z(R) \approx 2\pi G \Delta \rho (A_{\text{top}} + A_{\text{bot}}) \int_0^\infty J_0(kR) e^{-kz_0} \frac{\pi w^2}{2} \frac{k^2 w}{\sinh\left(\frac{\pi k w}{2}\right)} dk.$$

4.1 Effective-depth approximation and FWHM–depth estimator

For moderate-to-high k ,

$$\frac{1}{\sinh\left(\frac{\pi k w}{2}\right)} \approx 2 e^{-\frac{\pi k w}{2}},$$

so the combined damping $e^{-kz_0} / \sinh(\cdot)$ behaves like $e^{-kz_{\text{eff}}}$ with

$$z_{\text{eff}} \equiv z_0 + \frac{\pi}{2} w.$$

Matching leading moments yields the proxy

$$g_z^{\text{sig}}(R) \approx \mathcal{A} \frac{z_{\text{eff}}}{(R^2 + z_{\text{eff}}^2)^{3/2}}, \quad \mathcal{A} = C G \Delta \rho (A_{\text{top}} + A_{\text{bot}}) w^3,$$

with $C = \mathcal{O}(10)$ (a convenient peak-matched choice is $C \approx 4\pi^2$).

FWHM–depth relation. For this proxy,

$$R_{1/2} = z_{\text{eff}} \sqrt{2^{2/3} - 1} \approx 0.766 z_{\text{eff}}, \quad \text{FWHM} \approx 2R_{1/2} \approx 1.532 z_{\text{eff}}.$$

Therefore,

$$z_0 \approx \frac{\text{FWHM}}{1.532} - \frac{\pi}{2} w$$

providing a practical depth estimate once w is known (or co-estimated).

5. Comparison with a Buried Sphere

A sphere of radius a with center depth z_0 has

$$g_z^{\text{sphere}}(R) = \frac{4\pi G}{3} \Delta \rho a^3 \frac{z_0}{(R^2 + z_0^2)^{3/2}}.$$

The sigmoid-lens proxy shares this shape with z_0 replaced by z_{eff} , explaining why the two anomalies can be difficult to distinguish from profile data alone.

References

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